THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** infinitely many solutions (coincident lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
2	**(d)** 5	Solve by elimination
3	**(c)** 2	Solve by elimination
4	**(d)** 24	$x = 6$, $y = 4$
5	**(d)** 4	$\frac{1}{2} = \frac{2}{k} \neq \frac{3}{7}$
6	**(a)** 2	Solve, then add
7	**(d)** no solution (parallel lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
8	**(d)** 4	Solve by elimination
9	**(c)** 1	Solve by elimination
10	**(b)** 45	$x = 9$, $y = 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let father x, daughter y now (1); $x - 7 = 7(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1
12	Express one variable, substitute: $x = 5$ (1), $y = 3$ (1).	1+1
13	$\frac{1}{2} = \frac{2}{k} \neq \frac{3}{7}$ (1); $k = 4$ (1).	1+1
14	Let father x, daughter y now (1); $x - 7 = 6(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{a_1}{a_2} = \frac{5}{7}$, $\frac{b_1}{b_2} = -\frac{2}{3}$, $\frac{c_1}{c_2} = \frac{8}{9}$ (2); ratios $a_1/a_2 \neq b_1/b_2$: lines intersect (1).	2+1
16	Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x = -21$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
17	Let tens digit x, ones digit y: $x + y = 9$ (1); $(10y + x) - (10x + y) = -45$ so $y - x = -5$ (1); $x = 7$, $y = 2$, number 72 (1).	1+1+1
	OR: $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$ (2); all ratios equal: lines coincide (1).	2+1
18	Multiply (i) by -3 and (ii) by 2 and subtract: $(-13)x = -52$ (2); $x = 4$ (1); put in (i): $y = 1$ (1); check in (ii) (1).	2+1+1+1
	OR: Let tens digit x, ones digit y: $x + y = 9$ (1); $(10y + x) - (10x + y) = -27$ so $y - x = -3$ (1); $x = 6$, $y = 3$, number 63 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Son x, father $8x$ (1); $8x + 9 = 5(x + 9)$ (2); $x = 12$ (1); son 12 years, father 96 years (1).	1+2+1+1
	OR: Fixed x, per km y: $x + 10y = 170$, $x + 15y = 240$ (2); subtract: $5y = 70$, $y = 14$ (1); $x = 30$ (1); fare for 25 km $= 30 + 14 \times 25 = 380$ (1).	2+1+1+1
20	Son x, father $7x$ (1); $7x + 6 = 4(x + 6)$ (2); $x = 6$ (1); son 6 years, father 42 years (1).	1+2+1+1
	OR: Fixed x, per km y: $x + 8y = 141$, $x + 14y = 213$ (2); subtract: $6y = 72$, $y = 12$ (1); $x = 45$ (1); fare for 30 km $= 45 + 12 \times 30 = 405$ (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 9	$\frac{1}{3} = \frac{3}{k} \neq \frac{4}{13}$
2	**(c)** 7	Solve, then add
3	**(a)** exactly one solution (intersecting lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
4	**(a)** 3	Solve by elimination
5	**(a)** 2	Solve by elimination
6	**(b)** 55	$x = 11$, $y = 5$
7	**(c)** 8	$\frac{1}{4} = \frac{2}{k} \neq \frac{5}{21}$
8	**(b)** 11	Solve, then add
9	**(a)** no solution (parallel lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
10	**(a)** 2	Solve by elimination

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Express one variable, substitute: $x = 4$ (1), $y = 2$ (1).	1+1
12	$\frac{1}{3} = \frac{3}{k} \neq \frac{4}{13}$ (1); $k = 9$ (1).	1+1
13	Let father x, daughter y now (1); $x - 7 = 5(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1
14	Express one variable, substitute: $x = 3$ (1), $y = 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{a_1}{a_2} = 3$, $\frac{b_1}{b_2} = 3$, $\frac{c_1}{c_2} = \frac{10}{9}$ (2); $a_1/a_2 = b_1/b_2 \neq c_1/c_2$: lines are parallel (1).	2+1
16	Multiply (i) by 2 and (ii) by 3 and subtract: $(-1)x = -3$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
17	Let tens digit x, ones digit y: $x + y = 9$ (1); $(10y + x) - (10x + y) = -63$ so $y - x = -7$ (1); $x = 8$, $y = 1$, number 81 (1).	1+1+1
	OR: $\frac{a_1}{a_2} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{1}{3}$, $\frac{c_1}{c_2} = \frac{1}{3}$ (2); all ratios equal: lines coincide (1).	2+1
18	Multiply (i) by 4 and (ii) by 2 and subtract: $(14)x = 28$ (2); $x = 2$ (1); put in (i): $y = 5$ (1); check in (ii) (1).	2+1+1+1
	OR: Let tens digit x, ones digit y: $x + y = 9$ (1); $(10y + x) - (10x + y) = -9$ so $y - x = -1$ (1); $x = 5$, $y = 4$, number 54 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let boat x km/h, stream y km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $30p + 45q = 11$, $40p + 27q = 11$ (2); $p = \frac{1}{5}$, $q = \frac{1}{9}$ (1); $x - y = 5$, $x + y = 9$ (1); boat 7 km/h, stream 2 km/h (1).	2+1+1+1
	OR: Length x, breadth y: $(x - 5)(y + 3) = xy - 30$ gives $3x - 5y = -15$ (1.5); $(x + 3)(y + 2) = xy + 91$ gives $2x + 3y = 85$ (1.5); solving, $x = 20$, $y = 15$ (2).	1.5+1.5+2
20	Let boat x km/h, stream y km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $28p + 30q = 10$, $21p + 50q = \frac{41}{4}$ (2); $p = \frac{1}{4}$, $q = \frac{1}{10}$ (1); $x - y = 4$, $x + y = 10$ (1); boat 7 km/h, stream 3 km/h (1).	2+1+1+1
	OR: Length x, breadth y: $(x - 4)(y + 2) = xy - 20$ gives $2x - 4y = -12$ (1.5); $(x + 2)(y + 3) = xy + 84$ gives $3x + 2y = 78$ (1.5); solving, $x = 18$, $y = 12$ (2).	1.5+1.5+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	Solve by elimination
2	**(c)** 96	$x = 12, y = 8$
3	**(c)** 8	$\frac{1}{2} = \frac{4}{k} \neq \frac{3}{7}$
4	**(a)** 11	Solve, then add
5	**(b)** infinitely many solutions (coincident lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
6	**(d)** 6	Solve by elimination
7	**(d)** 2	Solve by elimination
8	**(d)** 128	$x = 16, y = 8$
9	**(c)** 15	$\frac{1}{3} = \frac{5}{k} \neq \frac{2}{7}$
10	**(d)** 5	Solve, then add

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{4} = \frac{2}{k} \neq \frac{5}{21}$ (1); $k = 8$ (1).	1+1
12	Let father x, daughter y now (1); $x - 7 = 8(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1
13	Express one variable, substitute: $x = 2$ (1), $y = 5$ (1).	1+1
14	$\frac{1}{2} = \frac{4}{k} \neq \frac{3}{7}$ (1); $k = 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{2}{5}$ (2); $a_1/a_2 = b_1/b_2 \neq c_1/c_2$: lines are parallel (1).	2+1
16	Multiply (i) by 2 and (ii) by -3 and subtract: $(13)x = 78$ (2); $x = 6$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
17	Let tens digit x, ones digit y: $x + y = 11$ (1); $(10y + x) - (10x + y) = -63$ so $y - x = -7$ (1); $x = 9$, $y = 2$, number 92 (1).	1+1+1
	OR: $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$ (2); all ratios equal: lines coincide (1).	2+1
18	Multiply (i) by -2 and (ii) by 4 and subtract: $(-26)x = -104$ (2); $x = 4$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
	OR: Let tens digit x, ones digit y: $x + y = 11$ (1); $(10y + x) - (10x + y) = -27$ so $y - x = -3$ (1); $x = 7$, $y = 4$, number 74 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let fraction $\frac{x}{y}$: $3(x + 1) = 2(y + 1)$ and $11(x + 3) = 8(y + 3)$ (2); simplify and solve (2); $x = 5$, $y = 8$, fraction $\frac{5}{8}$ (1).	2+2+1
	OR: Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $240p + 240q = 7$, $300p + 180q = \frac{29}{4}$ (2); $p = \frac{1}{60}$, $q = \frac{1}{80}$ (2); train 60 km/h, bus 80 km/h (1).	2+2+1
20	Let fraction $\frac{x}{y}$: $11(x + 2) = 6(y + 2)$ and $14(x + 5) = 9(y + 5)$ (2); simplify and solve (2); $x = 4$, $y = 9$, fraction $\frac{4}{9}$ (1).	2+2+1
	OR: Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $160p + 200q = 8$, $200p + 160q = \frac{41}{5}$ (2); $p = \frac{1}{40}$, $q = \frac{1}{50}$ (2); train 40 km/h, bus 50 km/h (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.